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Research paper

Adaptive kinematic and stiffness control of a hybrid-driven soft robot for enhanced interactions under external loads

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A R T I C L E I N F O

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A B S T R A C T

Soft robots offer promising solutions for human-robot interaction through inherent compliance, yet achieving controllable stiffness while maintaining dexterity remains challenging. This paper introduces a novel hybrid-actuated soft manipulator that exploits the synergistic interaction between tendon-driven mechanisms and pneumatic chambers to realize continuous stiffness modulation over an extensive range. A key challenge for this system is the precise prediction of the minimum actuation pressure required across diverse configurations, which constitutes a highly nonlinear modeling problem. We address this through a Weighted Broad Learning System (WBLS) that learns these complex pressure-configuration relationships with enhanced noise tolerance. Our primary contribution lies in developing an integrated control architecture that combines the WBLS-based pressure predictor with piecewise constant curvature (PCC) kinematics to enable coordinated regulation of both robot pose and mechanical stiffness. The system incorporates a feedback-driven adaptation mechanism that monitors end-effector responses to external disturbances and dynamically compensates for load-induced deformations by adjusting robot pose and stiffness parameters to maintain desired performance. Experimental validation confirms significant stiffness adjustability, achieving more than 3.6-fold variation in axial stiffness. Under external loading conditions, the proposed approach achieves comparable trajectory precision with a reduction in average driving pressure of up to 31.83% compared to conventional rigid-stiffness strategies. Practical validation through complex object handling tasks confirms the system's effectiveness in adaptive manipulation scenarios.

1. Introduction

Soft robots are constructed using materials with low Young's modulus [1], which enables them to absorb contact forces through substantial material deformations [2,3]. This characteristic offers significant potential for facilitating safe physical interactions between humans and robots, particularly in applications such as disability assistance during bathing [4], food collection tasks [5], and medical swab sampling [6]. In these contexts, manipulators must possess sufficient softness to minimize the risk of bodily harm during collisions, especially under constrained impact conditions. Simultaneously, these manipulators must provide adequate rigidity to perform essential physical operations including grasping and manipulation tasks [7]. Excessive softness can result in unanticipated deformations during physical interactions, thereby increasing the complexity of robot control. This necessitates the development of soft robots with variable stiffness capabilities for safe human-robot interactions and delicate object handling. These robots must demonstrate the ability to provide appropriate stiffness levels when subjected to high interaction forces.

Researchers have developed various techniques to induce stiffness changes, primarily employing material phase change and structural interference methods. The material phase change approach involves incorporating variable stiffness materials such as thermally sensitive functional materials [8], magnetorheological or electro-rheological fluids [9], and low-melting-point alloys [10,11] into soft robotic structures. The application of external thermal or electric fields triggers material transitions from solid to liquid states, thereby achieving wide stiffness variation ranges. However, this method typically provides only binary switching between rigid and flexible states, presenting difficulties in achieving continuous control and exhibiting slow response characteris-

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Nomenclature

Kinematic Parameters

$X(t)$	End-effector position	(m)
θ_k	Bending angle, segment k	(rad)
γ_k	Length, segment k	(m)
J	Kinematic Jacobian matrix	
$q(t)$	Configuration space parameters	
ϕ_k	Deflection angle, segment k	(rad)
l_{ki}	Tendon i length, segment k	(m)
h	Cross-section radius	(m)

WBLS Network Parameters

$P_{\min}$	Minimum driving pressure	(Pa)
λ	Ridge regression parameter	
n	Number of feature nodes	
Z_i	Output of i-th feature node	
W	Connection weight matrix	
σ	Weighted penalty vector	
m	Number of enhancement nodes	
H_j	Output of j-th enhancement node	

Stiffness Control Parameters

P	Chamber pressure	(Pa)
F_{ext}	External force	(N)
K_q	Joint stiffness	(N·m/rad)
$k_{\text{compression}}$	Axial stiffness	(N/m)
ΔX_n	Position error, moment n	(m)
ν	Stiffness regularization	
ΔP	Pressure increment	(Pa)
K_X	Cartesian stiffness	(N/m)
k_{bending}	Bending stiffness	(N·m/rad)
$U(P)$	Elastic potential energy	(J)
η	Proportional gain	
J^+	Jacobian pseudoinverse	

tics. Structural interference methods include particle jamming [12,13], fiber jamming [14], lockjaw jamming [15], layer jamming [16], and chain jamming [17] techniques. These approaches involve packing specific jamming structures within vacuum-sealed enclosures. Negative pressure within the enclosure causes jamming structures to interlock, preventing sliding motion and creating high-stiffness solid-like states while maintaining fluid-like properties at atmospheric pressure. However, the integration of jamming mechanisms into soft robot structures frequently increases system weight, which subsequently affects deformation characteristics. Furthermore, unpredictable structural arrangements reduce system repeatability and operational stability.

In contrast to the aforementioned methods, including structural complexity and controllability challenges, hybrid-driven approaches can achieve balanced performance combining simultaneous dexterity and stiffness control by emulating biological structures such as octopus arms [18]. Doi et al. [19] developed a bio-inspired soft robot that emulates octopus arm musculature using 32 thin McKibben actuators. These actuators enable contraction, bending, torsion, and stiffness modulation through selective actuation in axial, radial, and tilting directions. Kang et al. [20] designed a tendon-enhanced pneumatic continuum robot to improve positioning accuracy while maintaining large displacement capabilities. Zhang et al. [21] implemented non-stretchable origami air chambers that function as virtual skeletons, enabling 3D motion with variable stiffness over extensive ranges through antagonistic interactions between tendons and pneumatic pressure. These designs demonstrate effective variable stiffness capabilities with real-time physical stiffness adjustment functionalities.

The fundamental challenge in hybrid-driven systems lies in the inherent actuation-stiffness coupling phenomenon, where control inputs simultaneously govern both geometric deformation and mechanical compliance properties. Model-based approaches have demonstrated promise in addressing these challenges through physics-informed computational frameworks. Fairchild et al. [22] developed sophisticated estimation algorithms that integrate Euler-Bernoulli beam mechanics with Bayesian filtering techniques to achieve accurate online stiffness prediction using only limited positional sensor data. Their approach successfully demonstrated real-time parameter adaptation capabilities under varying payload conditions and robot configurations. Similarly, Zhang et al. [23] proposed advanced deformation feedforward control strategies specifically designed for underwater soft robotic applications. However, these model-based approaches remain fundamentally constrained by their dependence on precise a priori system identification and comprehensive understanding of material properties. Consequently, they exhibit significant performance degradation when confronted with unmodeled dynamics, material parameter uncertainties, manufacturing tolerances, and time-varying external disturbances.

Data-driven methodologies have emerged as promising alternative solutions to address the inherent limitations of explicit modeling approaches. Advanced reinforcement learning techniques have shown considerable potential in this domain. Thuruthel et al. [24] implemented sophisticated model-based policy gradient algorithms to derive adaptive closed-loop control policies that demonstrate superior load accommodation compared to conventional approaches. Meanwhile, Zhu et al. [25] leveraged deep deterministic policy gradient (DDPG) algorithms for achieving real-time stiffness modulation in variable-stiffness grippers during high-speed grasping operations. Their experimental results illustrate the substantial capability of learning-based approaches to handle complex nonlinear mappings without requiring explicit system models. Nevertheless, reinforcement learning deployment encounters practical barriers, including extensive training data requirements, limited sample efficiency, sensitivity to hyperparameter selection, and computational overhead that constrains real-time applicability in resource-constrained robotic systems.

To enable robust and efficient control of hybrid-driven soft robots, it is essential to develop computational models that can accurately capture the complex nonlinear relationships between actuation and stiffness while maintaining computational tractability [26,27]. Such models must provide a reliable computational foundation that can support simultaneous optimization of both kinematic trajectories and stiffness profiles under dynamic operating conditions. Recent advances in machine learning have demonstrated exceptional capability in modeling complex, nonlinear system dynamics without explicit mathematical derivation [28–30]. Neural network-based methods have achieved notable success in environmental prediction, time-series analysis, and spatiotemporal modeling using deep architectural frameworks [31,32]. However, their deployment in real-time soft robot control encounters significant limitations including overfitting with limited training data, sensitivity to sensor noise, substantial computational requirements, and convergence challenges during online adaptation processes. The Weighted Broad Learning System (WBLS) [33] provides efficient nonlinear mapping capabilities with built-in robustness to anomalous data and computational efficiency suitable for real-time deployment scenarios.

This paper presents a hybrid-driven soft manipulator with extensive workspace and continuous stiffness variation capabilities. In soft robotic systems, the integration of tendon and pneumatic actuation mechanisms plays a crucial role in regulating end-effector positions and ensuring structural stability. The tendon-driven component pri-

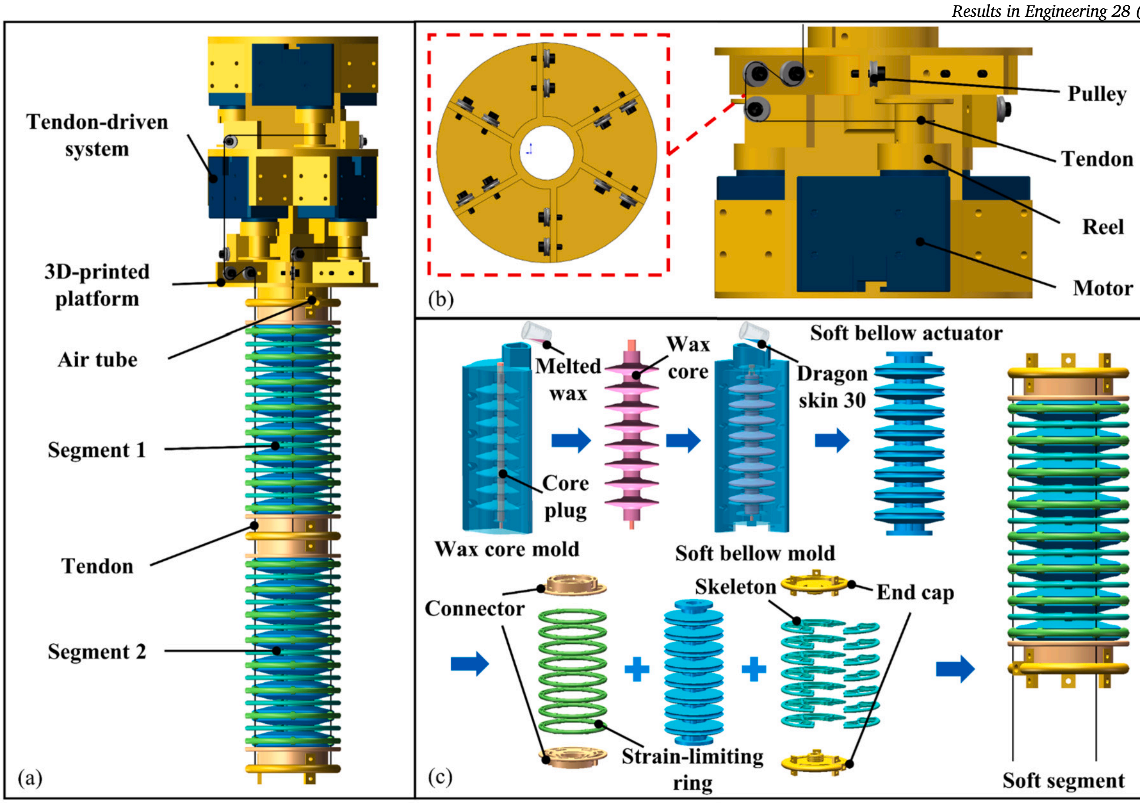

Fig. 1. Architecture of the hybrid-driven origami continuum robot. (a) Layout of the soft manipulator and prototype assembly. (b) Tendon drive system. (c) The fabrication process and assembly of the soft segment.

marily adjusts end-effector positioning, while the pneumatic actuation system initially achieves minimum air pressure for structural integrity before enabling variable stiffness modulation. We employ the WBLS framework to establish nonlinear mappings between minimum driving pressures and system configurations, while effectively managing sensor noise and measurement uncertainties with minimal computational overhead. Furthermore, the combination of WBLS and the PCC model forms the comprehensive kinematic model for no-load soft manipulator operation. For end-loading scenarios, an inverse kinematic model is carefully developed to optimize both configuration and stiffness parameters simultaneously. This model accounts for deviations caused by external forces and inherent stiffness variations. Additionally, real-time stiffness estimation is refined through feedback monitoring of end-effector error. Comprehensive experimental validations highlight the hybrid-driven soft robot's proficiency at dynamically adjusting stiffness across varying operational conditions and load profiles.

The remainder of this paper is organized as follows: Section 2 describes the manipulator design and fabrication process. Sections 3 and 4 develop the WBLS-PCC kinematic model under no-load conditions and establish pressure-configuration mappings. Section 5 presents the load-aware kinematic control model. Section 6 validates the integrated system through comprehensive experiments. Section 7 concludes the paper.

2. Design and fabrication

2.1. Robot design

Fig.1 illustrates the proposed soft manipulator, comprising two soft driven modules, a tendon drive system, and a 3D-printed base. Each module consists of a bellow chamber and reinforcement skeletons. The silicone bellow chamber exhibits elongation under positive internal pressure, enabling a wide range of movement and supporting the structure. Strategically placed rigid skeletons at the peaks and valleys of the chamber restrict radial expansion and enhance overall structural stiffness. The distal segment's skeleton contains three equally spaced tendon channels, while the proximal has six. The soft robot is driven by six individually controlled tendons via the tendon drive system at the base (Fig. 1(c)). This system comprises two layers, each with three motors and corresponding winding wheels. The configuration allows each set of motors and wheels to govern the movement of a single tendon, thereby facilitating the soft segment's three degrees of freedom through precise adjustments of the tendon lengths. The deformations exhibited by the segment, specifically bending and telescoping, are predominantly dictated by the lengths of the tendons engaged, with contraction inducing passive morphological alterations in the bellow chamber. Tension within the tendons is maintained via pneumatic pressure, and modifications to this input pressure effectuate changes in the internal forces and the stiffness of the module. Thus, this hybrid driven system proactively modulates both the motion and the stiffness of the robot. The capability to adjust stiffness is imperative for continuum robots, as it significantly enhances both manipulation precision and load-bearing capacity. The proposed design ensures the extension capability of the continuum robot and improves manipulation performance.

2.1.1. Manufacturing and assembly

The fabrication of the soft bellow chamber for each soft driven module was executed using silicone molding techniques, as illustrated in Fig. 1(c). The process commenced with the production of a wax core, employing 3D printing technology for mold creation. Subsequently, the constructed mold was filled with melted wax, and a stiff core plug was added to aid the fixing process thereafter. Upon solidification, the wax core was carefully extracted and subjected to further processing to attain a smooth, burr-free finish. This prepared wax core was then assembled with 3D-printed rigid outer molds. To facilitate seamless demolding of the soft bellow actuator, a silicone release agent (Release 300, Smooth-on Inc.) was uniformly applied to all mold surfaces. Following this, silicone rubber (Dragon Skin 30, Smooth-On Inc.) was injected into the

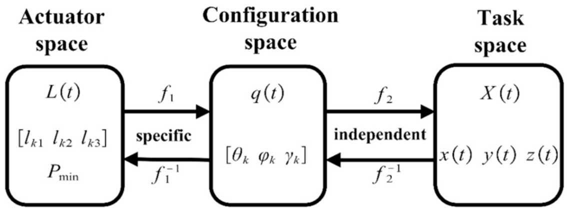

Fig. 2. The kinematic mapping for the soft robot.

molds and allowed to cure. The bellows containing the wax core were then heated to 70 °C to liquefy and remove the wax, leaving behind the hollow silicone structure. To limit the radial expansion of the bellows, external rigid strain-limiting rings were fixed at the peak using a silicone adhesive (Sil-Poxy, Smooth-On Inc.). Additionally, rigid skeletons were strategically positioned within the grooves of the bellows to provide support. The assembly was completed with the attachment of end caps and connectors to both ends of the manipulator.

For operational control, pneumatic pressure was regulated by a proportional valve (EPV2-50MD2, XINGYU) with a rapid 10 ms response time. This valve interfaces and operates through an RS485 bus, ensuring consistent pneumatic pressure for precise system operations. Six high-quality DC motors (XM430-W350-R, Dynamixel) are used to control the winding of the six tendons via 3D-printed winding wheels. To reduce friction and enhance tendon smoothness, three metallic rollers serve as pulleys. The assembly culminates with integrating two soft segments, the DC motors, winding wheels, rollers, and various standard components like bearings, bolts, and nuts, onto a specially designed 3D printed base platform, yielding a functional prototype.

3. WBLS-based kinematic model with no load

As previously indicated, air pressure facilitates the deformation of bellows, thereby creating a stretchable pneumatic virtual skeleton that enhances structural stability. Additionally, air pressure is employed to modulate stiffness by adjusting structural stress through tendon antagonism. To effectively decouple motion and stiffness regulation via pneumatic actuation, it is imperative to precisely define the minimum pressure required to sustain bellow deformation across various configurations. This relationship is delineated using the weighted broad learning system (WBLS) method. The deformation pose of the soft robot is characterized using the piecewise constant curvature (PCC) model. A typical PCC model necessitates a mapping among actuator, configuration, and task spaces as illustrated in Fig. 2. The task space encompasses end-effector position components, denoted as $X(t) = [x(t), y(t), z(t)]^T$, whilst the configuration space $q(t)$ describes the deformation parameters of the soft manipulator. The parameters of segment k can be described as bending angle θ_k, deflection angle ϕ_k, and length γ_k. Actuator space consists of tendon length $[l_{k1}, l_{k2}, l_{k3}]$ and pressure $P_{\min}$.

3.1. Weighted broad learning system

The WBLS, as described in [33], dynamically assigns weights to samples in a dataset. Samples with high reliability receive increased weights, while potential outliers are assigned reduced weights. This strategic weighting mitigates the impact of noise and outliers on model performance. Additionally, the WBLS has a shorter training duration than more complex architectures like deep neural networks, enabling faster deployment. The structure of the network is shown in Fig. 3. In the architecture of the neural network, the input nodes undergo an initial transformation into feature nodes, which are subsequently converted into enhancement nodes through the application of random weights. These mapped features and enhancement nodes establish a linkage with the output layer. The connection weights within the neural network are determined using the ridge regression approximation algorithm.

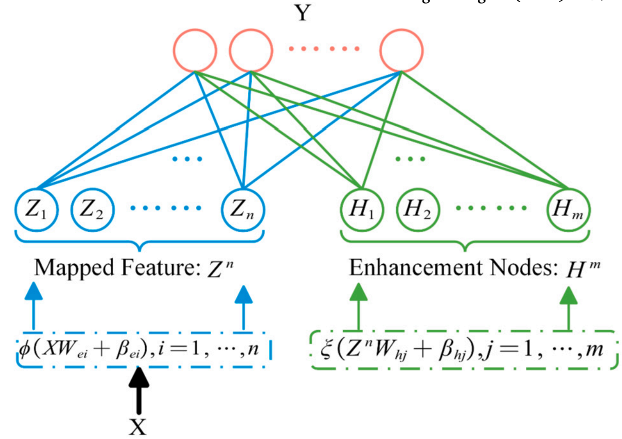

Fig. 3. Illustration of WBLS.

Consider the training dataset $\{x_i, y_i\}_{i=1}^N$, $x_i = [\theta_i, \phi_i]^T$ is the configuration in which the soft manipulator is placed, $y_i = [P^i_{\min}]^T$ is the minimum driving pressure. The network has n feature nodes and m enhancement nodes. The output of the ith feature node is:

$$Z_i = \phi(XW_n + \beta_n), \quad i = 1, \dots, n. \tag{1}$$

The output of jth enhancement node is:

$$H_j = \xi(Z^n W_m + \beta_m), \quad j = 1, \dots, m \tag{2}$$

where $Z^n = [Z_1, \dots, Z_n]$, $H^m = [H_1, \dots, H_m]$, W_{ei}, β_{ei}, W_{hi} and β_{hi} randomly generated from a given range of appropriate distributions $[-\zeta, \zeta]$. ϕ and ξ are some nonlinear activation functions. We set $A = [Z^n | H^m]$, then, the final broad learning network can be represented as follows:

$$Y = AW \tag{3}$$

The connection weights for WBLS can be calculated by the weighted ridge regression algorithm as follows:

$$\underset{W}{\arg\min} : \|\sigma AW - \sigma Y\|_2^2 + \lambda \|W\|_2^2 \tag{4}$$

where $\sigma = [\sigma_1, \sigma_2, \dots, \sigma_N]^T$ represents a weighted penalty factor that significantly influences the sample's weight. This factor is calculated using the Huber weight function, accommodating outliers by minimizing the impact of large residuals through a weighting mechanism. The Huber weight function is mathematically represented as follows:

$$\sigma_i = \begin{cases} 1, & |u_i| \le b \\ b/|u_i|, & |u_i| > b \end{cases} \tag{5}$$

where σ_i denotes ith weighted penalty factor; b denoted a positive adjustable parameter; u_i is the standardized residual of the ith sample obtained as:

$$u_i = \frac{r_i}{\hat{z}} \tag{6}$$

where r_i represents the discrepancy between the predicted values and the observed measurements for the ith sample. These discrepancies, referred to as residuals, collectively constitute the residual matrix $r = [r_1, \dots, r_N]^T$; $\hat{z}$ is a robust estimate of the standard deviation, which is estimated using the median absolute deviation with the following formula:

$$\hat{z} = 1.4826\text{MAD} = 1.4826\,\text{med}(|r_i - \text{med}(r)|) \tag{7}$$

where $\text{med}(\cdot)$ is the median function.

Then the connection weights of WBLS can be calculated by the following equation:

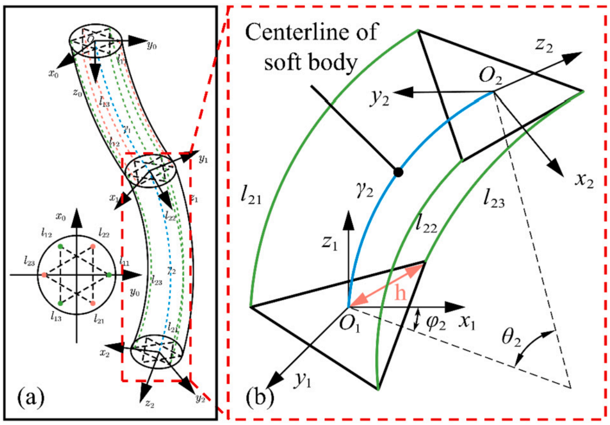

Fig. 4. Kinematic model of the soft robot.

$$W = (\lambda I + A^T \sigma^2 A)^{-1} A^T \sigma^2 Y \tag{8}$$

In the algorithm, the connection weights and weighted penalty factors can be calculated iteratively for better accuracy. Algorithm 1 outlines the procedure for network training.

Algorithm 1: Weighted Broad Learning System.

Data: Training data X
Result: W
for $i = 1; i \leqslant n; i++$ **do**
  Random W_{ei}, β_{ei};
  Calculate $Z_i = \phi\left(XW_{ei} + \beta_{ei}\right)$
Set the feature mapping group $Z^n = \left[Z_1, \cdots, Z_n\right]$;
for $j = 1; j \leqslant m; j++$ **do**
  Random W_{hj}, β_{hj};
  Calculate $H_j = \xi\left(Z^n W_{Hj} + \beta_{Hj}\right)$
Set the enhancement node group $H^n = \left[H_1, \cdots, H_m\right]$;
Set $A^m = [Z^n | H^m]$ and calculate residual error;
Let initial σ be an identity matrix;
if *stopping iterative criterion is not satisfied* **then**
  Update σ;
  Calculate W_m, A_m^+ with Eq.;
Set $W = W_m$;

Then, the weight matrix W is obtained from Algorithm 1. Given the weight W, we can predict $P^*_{\min}$ from new input $X = [\theta^*, \gamma^*]$ by:

$$P^*_{\min} = f_{\text{WBLS}}(X) \tag{9}$$

where $f_{\text{WBLS}}(X) = [Z_1^*, \cdots, Z_n^*, H_1^*, \cdots, H_m^*]W$, Z_i^*, $i = 1, \cdots, n$ can calculate by (1). H_j^*, $j = 1, \cdots, m$ can calculate by (2).

3.2. Kinematic model

3.2.1. Robot-independent transformation

The robot-independent mapping connects the configuration space and the task space. As shown in Fig. 4, a local coordinate system is established at each segment base, with the origin at the manipulator base center point O and z-axis aligned with the manipulator axis. Through geometric analysis, this configuration yields specific position vectors for the terminal centers of segments.

$$p = \begin{bmatrix} \frac{\gamma_k c\varphi_k(1-c\theta_k)}{\theta_k} & \frac{\gamma_k s\varphi_k(1-c\theta_k)}{\theta_k} & \frac{\gamma_k s\theta_k}{\theta_k} \end{bmatrix}^T \tag{10}$$

where $c(\cdot)$ and $s(\cdot)$ are abbreviations for $\cos(\cdot)$ and $\sin(\cdot)$, respectively.

The orientation of the tip of the kth segment can be determined by three successive rotations: rotating φ_k around the z-axis, rotating θ_k around the y-axis, and rotating $-\varphi_k$ reversely around the z-axis of the current tip frame to ensure the torsion-less assumption within each segment. Hence, the rotation matrix of the kth segment can be expressed by represented by the matrix shown below:

$$\begin{aligned} R &= \begin{bmatrix} R_{z,\varphi_k} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{y,\theta_k} & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} R_{x,-\varphi_k} & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 + c^2\varphi_k\left(c\theta_k - 1\right) & s\varphi_k c\varphi_k\left(c\theta_k - 1\right) & c\varphi_k s\theta_k \\ c\varphi_k s\theta_k\left(c\theta_k - 1\right) & 1 + s^2\varphi_k\left(c\theta_k - 1\right) & s\varphi_k s\theta_k \\ -c\varphi_k s\theta_k & -s\varphi_k s\theta_k & c\theta_k \end{bmatrix} \end{aligned} \tag{11}$$

By collocating the position vector and the rotation matrix, the homogeneous transformation matrix from the original coordinate system to the new coordinate system can be obtained as follows:

$${}_{k-1}^{\;\;k}T = \begin{bmatrix} R_{3\times 3} & p_{3\times 1} \\ 0_{1\times 3} & 1 \end{bmatrix} \tag{12}$$

For a two-segment soft manipulator, the transition from configuration space to task space is as follows:

$${}_0^2T = {}_0^1T\,{}_1^2T \tag{13}$$

The Jacobian matrix is a multidimensional form of the partial derivatives with respect to forward kinematic time, reflecting the linear relationship between the rate of change of the variables in configuration space and the linear and angular velocities of the end coordinate system, defined as follows:

$$\dot{X}(t) = J(q(t))\dot{q}(t) \tag{14}$$

3.2.2. Robot-specific transformation

In the preceding discussion, we outlined the correlation between the configuration of a soft manipulator and the minimum chamber pressure required to maintain tension in all tendons. This relationship is critical as it maps tendon lengths to corresponding robot configurations. The full range of configuration variables can be expressed in terms of the tendon actuation parameters.

As shown in Fig. 4(b), considering the structure and size of the flexible arm and assuming that the tendon is not stretchable, the relationship from actuator space to configuration space can be described as:

$$q_k = \begin{bmatrix} \theta_k \\ \varphi_k \\ \gamma_k \end{bmatrix} = \begin{bmatrix} \frac{2\sqrt{l_{k1}^2 + l_{k2}^2 + l_{k3}^2 - l_{k1}l_{k2} - l_{k2}l_{k3} - l_{k1}l_{k3}}}{3h} \\ \operatorname{atan2}\left(\sqrt{3}\left(l_{k2} + l_{k3} - 2l_{k1}\right), 3\left(l_{k2} - l_{k3}\right)\right) \\ \frac{l_{k1} + l_{k2} + l_{k3}}{3} \end{bmatrix} \tag{15}$$

where h indicates the cross-section radius.

The relationship between the length of the robot tendon and the configuration is:

$$l_{ki} = \gamma_k - h\theta_k \cos\left(\delta_k + \frac{2\pi(i-1)}{3} - \varphi_k\right) \tag{16}$$

where $i \in \{1, 2, 3\}$ indicates the number of the tendon in the segment, $\delta_1 = \pi/2$, $\delta_2 = 5\pi/6$

4. Experiments and results of minimum driving pressure

The minimum driving pressure (P_{min}) refers to the lowest air pressure necessary to attain a particular configuration in a soft robot. To establish the relationship between minimum driving pressure and posture, we used a pneumatic system to drive the deformation of the soft robot in a tendon-locking situation. The current chamber air pressure was measured using an air pressure sensor (PSE532, SMC). Two marker points are mounted at each end of the soft driven module; their positions are recorded by an OptiTrack motion capture system. From these, the elongation length and bending angle of the module under different driving conditions are calculated. We consider the actuation pressure required to maintain tendon tension during module extension as the minimum pressure at which the actuation chamber length equals the

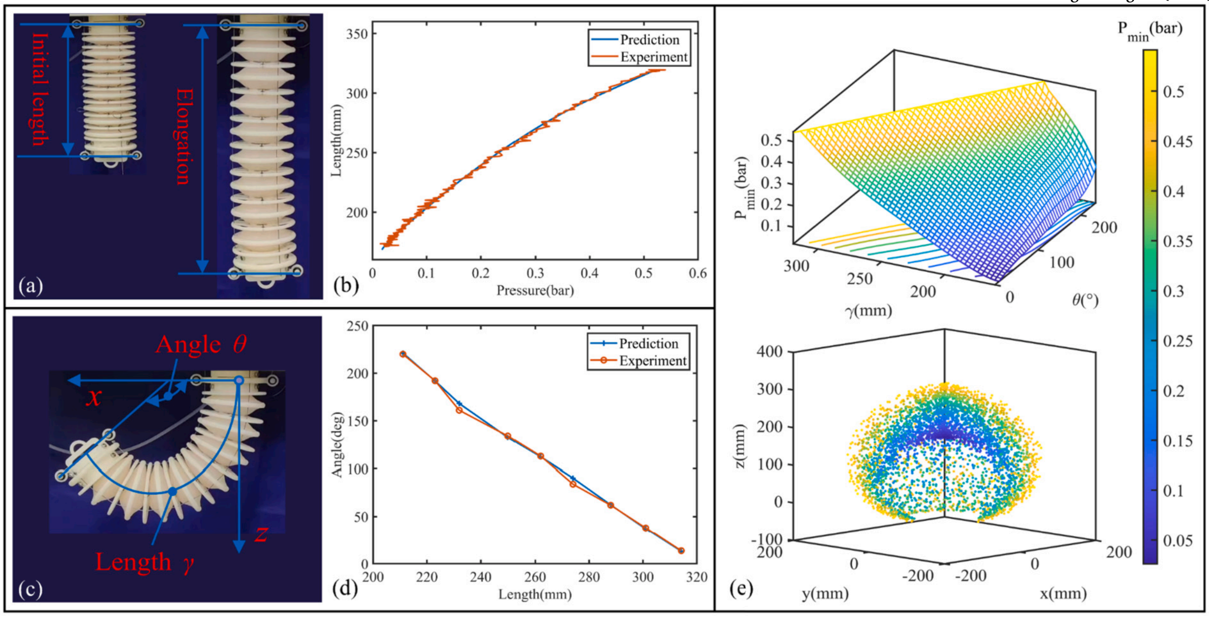

Fig. 5. (a) Elongation deformations. (b) Elongation of the soft driven module under different pressure. (c) Bending deformations. (d) Maximum bending angle at different elongation lengths. (e) Simulation of minimum driving pressure for soft driven module.

tendon length. For bending deflection, it is challenging to measure due to the differing effective lengths of three tendons and the chamber's passive deformation. We assume a fixed bending angle of the chamber, with a constant minimum pressure required across rotating angles. Therefore, only the minimum pressure for bending towards a single tendon, with the other two relaxed, needs to be measured. To simplify the driving system, we only consider positive pressure driving, setting the maximum minimum driving pressure to 0.5 bar. Before the experiments, we carefully calibrated the camera's working space; the error in the camera's stereo space was 0.08 mm.

4.1. Data collection

Two sets of experiments were conducted to characterize the designed soft driven module: uniaxial elongation and elongation-bending coupling. In the uniaxial elongation experiment (Fig. 5(a)), all tendons were relaxed. The initial module length was 166 mm, and the current chamber length was calculated from the two sets of marker points. During the experiment, the chamber was inflated to the maximum pressure of 0.5 bar, and then slowly deflated. Experimental data were recorded at 10 Hz. In the elongation-bending coupling motion (Fig. 5(b)), only one tendon length was restricted while the other two were relaxed. The module was driven by air pressure to bend in the direction of the restricting tendon. The limiting tendon length increased by 10 mm from the initial length up to maximum elongation. The inflation sampling process was the same as the uniaxial experiment. Since the module must be inflated and elongated to the same length as the limiting tendon before producing the corresponding bending motion, we only recorded data for bending angles greater than 0°.

4.2. Identification results

The data was initially normalized, followed by the application of the WBLS to accurately model experimental observations. The neural network was designed, trained, and validated using Python 3.8. Within the WBLS architecture, the parameter configuration critically influences the network's learning capabilities. An excessive number of nodes can lead to feature redundancy and increased computational demands. Conversely, insufficient nodes may fail to capture all necessary features, especially when processing extensive datasets. In this study, the feature

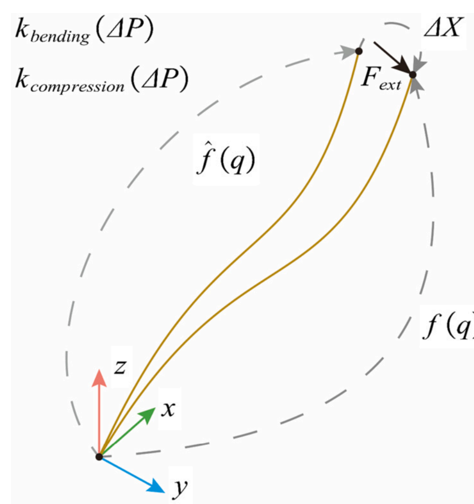

Fig. 6. The deformation curve of soft robots under external forces.

nodes and enhancement nodes of the network were configured to 10 and 20, respectively. The Sigmoid function facilitated nonlinear mapping within these nodes. Fig. 5(e) illustrates the trained neural network's efficacy in predicting minimal air pressure for single-axis elongation, demonstrating the WBLS network's adeptness at mitigating measurement noise effects. Additionally, the network predicted the maximum bending angle of the soft module with an average prediction error of 1.2° across varying lengths.

5. Motion control with external load via stiffness modulation

The inherent flexibility of soft robots, while beneficial for safe interaction, often leads to significant positional deviations under external loads (Fig. 6). The kinematic model presented in Section 3, designed for no-load conditions, lacks the capability to modulate stiffness in response to such deviations. This frequently results in excessive deformation and compromised control accuracy. To address this limitation, we extend the initial kinematic framework by developing motion control strategies that explicitly account for external loads and variable stiffness. We

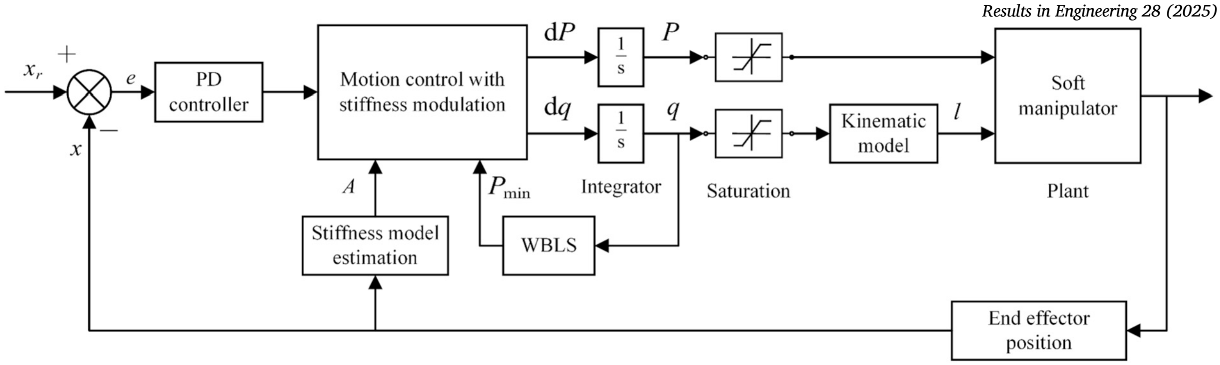

Fig. 7. Block diagram of the motion controller.

propose a novel control methodology that incorporates real-time stiffness adjustment to enhance both precision and adaptability in dynamic environments.

Under the assumption of negligible gravity and external forces, the kinematic mapping from configuration space to task space derived in Section 3 yields:

$$X = \hat{f}(q) \tag{17}$$

where $\hat{f}(\cdot)$ represents the configuration-to-task-space mapping. When external forces F_{ext} act on the system beyond what the initial stiffness can resist, the end-effector position deviates by $\mathrm{d}X$, leading to:

$$f(q) = X + \mathrm{d}X = \hat{f}(q) + K_X^{-1} F_{ext} \tag{18}$$

where K_X denotes the initial Cartesian stiffness of the soft robot.

Maintaining positional accuracy under external loading remains challenging for soft robots. Active stiffness modulation during motion provides an effective mechanism to mitigate model discrepancies and enhance robustness against perturbations. To establish the foundation for stiffness control, we investigate the relationship between joint-space and Cartesian-space stiffness, developing a systematic mapping from pneumatic pressure variations to Cartesian stiffness.

The configuration space stiffness, K_q, relates external forces to displacements in joint space. Following the derivation in [27], the general form is given by:

$$K_q = \frac{\mathrm{d}\left(J^T K_X \Delta X\right)}{\mathrm{d}q} = J^T K_X J - \frac{\mathrm{d}J}{\mathrm{d}q} K_X \Delta X \tag{19}$$

Under the assumption of static equilibrium and small deformations, the second term can be neglected, leading to the simplified and widely-used relationship:

$$K_q = J^T K_X J \tag{20}$$

For non-square Jacobian matrices, the Cartesian stiffness can be obtained through the pseudoinverse:

$$\begin{aligned} K_X &= J^{+T} K_q J^{+} \\ K_X^{-1} &= J K_q^{-1} J^T \end{aligned} \tag{21}$$

The bending and axial stiffness in joint space exhibit approximately linear relationships with pressure increment [26]:

$$\begin{aligned} k_{bending}(\Delta P) &= k^1_{P_{min}} + g^1_{P_{min}} \Delta P \\ k_{compression}(\Delta P) &= k^2_{P_{min}} + g^2_{P_{min}} \Delta P \end{aligned} \tag{22}$$

where $\Delta P = P - P_{min}$ represents the pressure increment beyond the minimum required for configuration maintenance. Parameters $k^1_{P_{min}}$ and $k^2_{P_{min}}$ denote the initial bending and axial stiffness at minimum pressure, while $g^1_{P_{min}}$ and $g^2_{P_{min}}$ are the corresponding pressure-stiffness coefficients.

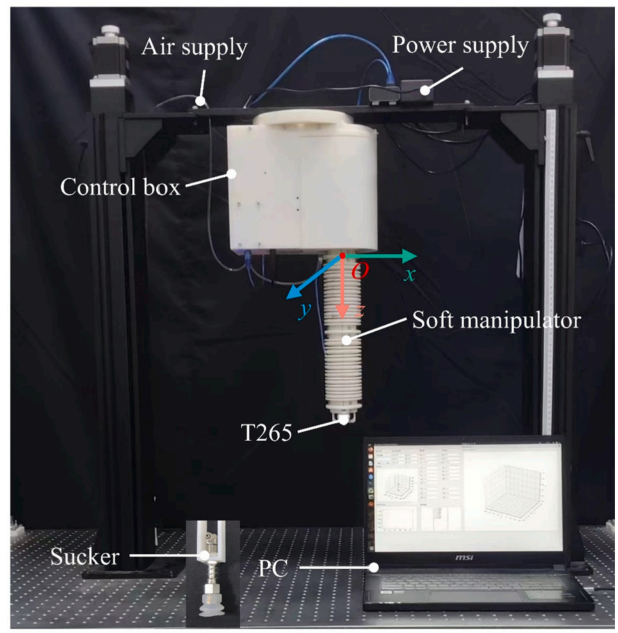

Fig. 8. Experiment setup.

The elastic potential energy stored in a single segment is formulated as:

$$U(P) = \frac{1}{2} k_{bending}(\Delta P)\theta^2 + \frac{1}{2} k_{compression}(\Delta P)\gamma^2 \tag{23}$$

The configuration space stiffness matrix is derived as the Hessian of the elastic potential energy:

$$\begin{aligned} K'_q &= \nabla^2 U = A\Delta P' \\ K_q &= \mathrm{diag}(K'_q) \\ K_q &\succ 0 \end{aligned} \tag{24}$$

where $\Delta P' = [1, \Delta P]^T$ is the augmented pressure vector.

To achieve precise tip positioning under external loads, we formulate an optimization problem that simultaneously determines optimal configuration and stiffness adjustments:

$$\begin{aligned} &\arg\min_{\mathrm{d}q,\mathrm{d}P} \quad \|\mathrm{d}X - J_q \mathrm{d}q\|_2^2 + \nu \|J_s [\mathrm{d}q \quad \mathrm{d}P^T]^T\|_2^2 \\ &\text{s.t.} \quad X_c = f(q) + K_X^{-1} F_{ext} \\ &\qquad K_X^{-1} = J(q) K_q^{-1} J^T(q) \\ &\qquad K_q = \mathrm{diag}(A\Delta P') \\ &\qquad P = P + \mathrm{d}P \\ &\qquad q = q + \mathrm{d}q \end{aligned} \tag{25}$$

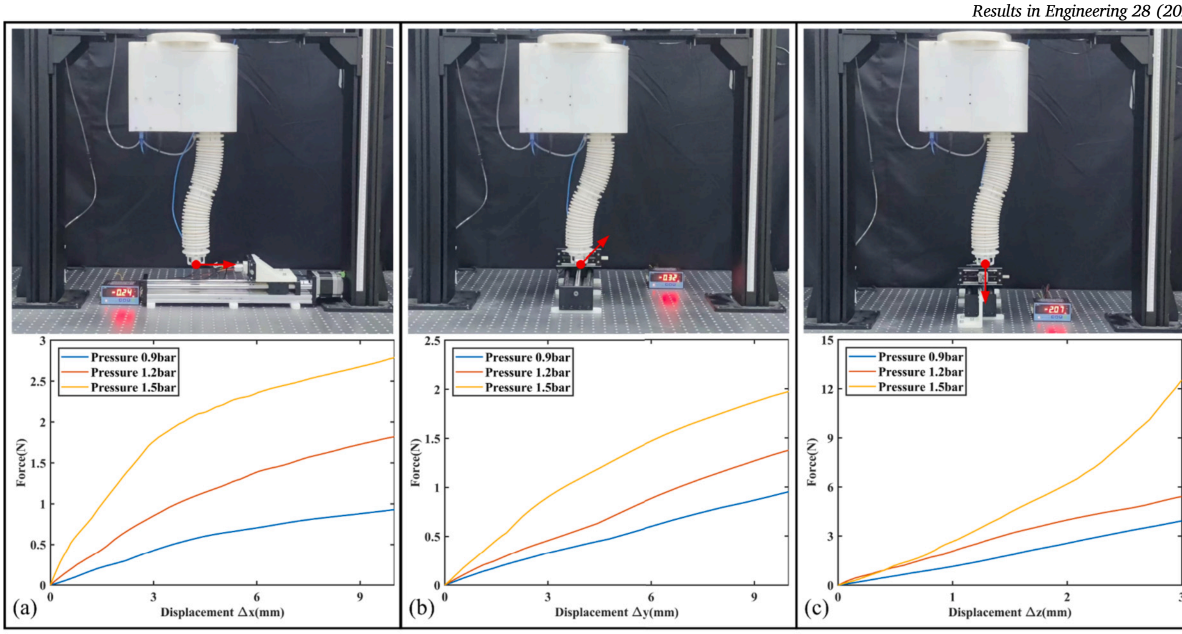

Fig. 9. Stiffness characterization tests of the soft manipulator. (a).

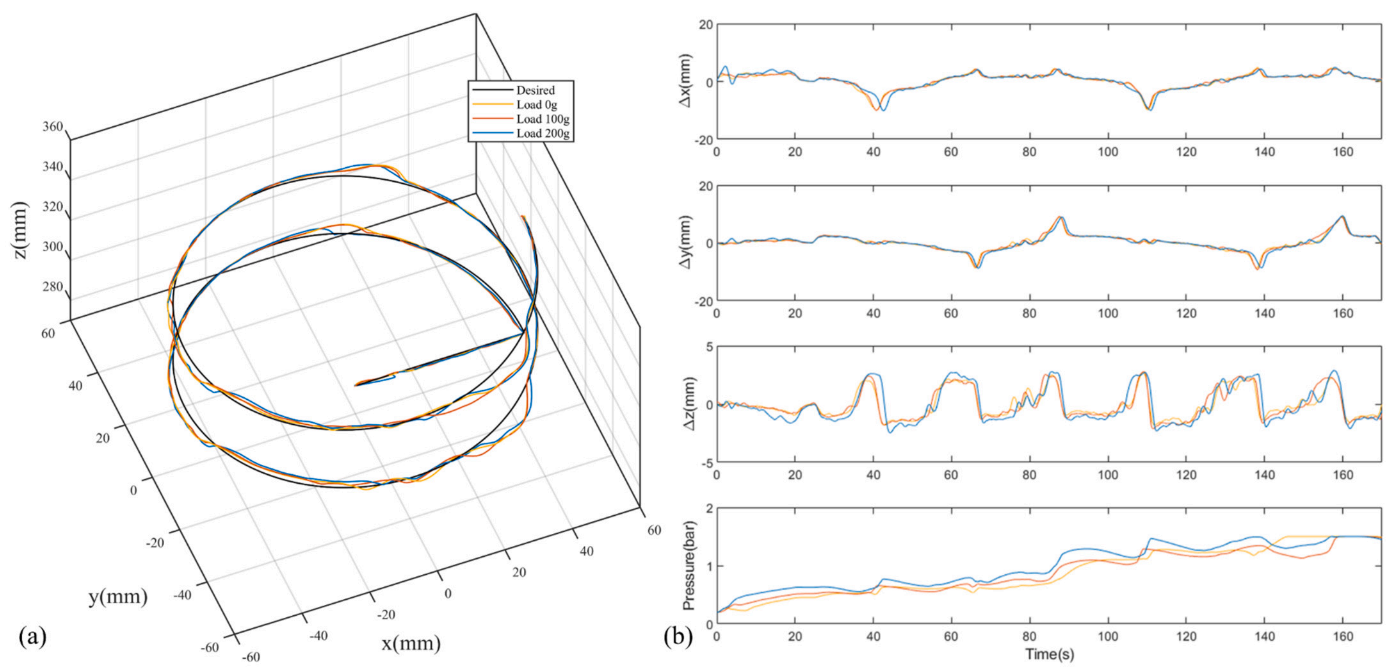

Fig. 10. Trajectory tracking performance under different external payload conditions. (a) Three-dimensional trajectory comparison showing the desired helical path (black line) and actual robot trajectories under various loading conditions. (b) Time-domain analysis of end-effector position components showing x, y, and z coordinates (top three panels) and orientation angle (bottom panel).

where J_q is the kinematic Jacobian, J_s is the sensitivity Jacobian of the stiffness model, and $\nu > 0$ is a weighting parameter that balances the trade-off between position accuracy and stiffness change. If $\nu = 0$, the problem reduces to standard inverse kinematics without stiffness compensation.

The highly nonlinear deformation characteristics of soft manipulators and the difficulty in directly measuring external forces present additional challenges. However, end-effector positioning errors are readily measurable, enabling force estimation through error normalization. The force direction is determined based on the error sign, and joint-space stiffness is adaptively adjusted using error feedback to compensate for load-induced deformations.

Let $\Delta X_n = X_d - X^n$ define the end-effector position error at the n-th control step, where X_d is the desired position and X^n is the current measured position. The joint-space stiffness is updated iteratively as:

$$K_q^{n+1} = K_q^n + \eta J_n^+ \Delta X_n \tag{26}$$

where K_q^{n+1} is the updated joint stiffness estimate, η is the proportional gain, and J_n^+ is the Moore-Penrose pseudoinverse of the Jacobian at step n.

Furthermore, the coefficients characterizing the stiffness-pressure relationship vary with both pose and current stiffness. We refine these estimates by solving the constrained optimization problem:

$$
\begin{aligned}
&\arg\min_{A} \quad \|\Delta A\|_2 \\
&\text{s.t.} \quad K_q' = A^{n+1}\Delta P' \\
&\quad A^{n+1} = A^n + \Delta A \\
&\quad K_q \succ 0
\end{aligned}
\tag{27}
$$

where $\|\Delta A\|_2$ represents the Frobenius norm of the parameter update matrix. Minimizing this norm ensures gradual variation in Jacobian estimates across time steps, leading to smoother controller performance.

The overall control architecture, integrating the WBLS-based pressure prediction, PCC kinematics, and adaptive stiffness regulation, is illustrated in Fig. 7.

6. Experiment and results

6.1. Experiment setup

The experimental setup involved a soft manipulator securely mounted on a metallic frame with its tip oriented vertically downward, ensuring unrestricted movement within the frame. The Robot Operating System (ROS2) was deployed on a laptop running Ubuntu 20.04 to enable seamless communication between the system and diverse sensors and controllers. Accurate localization of the manipulator's endpoint was achieved using an Intel RealSense tracking camera T265, mounted on the robot's end segment. The camera was connected to the laptop via USB 3.0 for rapid data transmission. As a safety precaution, the maximum pressure permissible through the proportional valve was restricted to 150 kPa throughout the duration of the experiment (Fig. 8).

6.2. Variable stiffness experiment

To assess the soft manipulator's capability for active stiffness modulation, an experimental investigation was conducted under three differentiated operational conditions. Fig. 9 depicts the experimental setup and corresponding outcomes. The manipulator was subjected to pneumatic pressures of 0.9 bar, 1.2 bar, and 1.5 bar, inducing varying antagonistic forces. Throughout the procedure, the manipulator was configured in an S-shape, characterized by the high rigidity of the robot. A tensiometer affixed to a uniaxial sliding table was employed to measure the force-displacement relationship. This apparatus was connected to the manipulator's terminal end via a tether, facilitating the acquisition of data for three axial force measurements. The experimental findings demonstrate that over a limited range of motion, the tension force exhibits nearly linear escalation with displacement, allowing the approximation of stiffness as a constant value. These results corroborate the hypothesis that the soft robot behaves analogously to a linear spring, where deformation under stress exhibits a direct linear correlation with stiffness coefficient over a limited deformation range. In the conducted experiments, the manipulator demonstrated notable axial compliance when subjected to a reduced driving pressure of 0.9 bar, achieving an axial displacement of approximately 0.65 N across a 4 mm span in the x-axis. This behavior can be attributed to the inherent compliance of the soft silicone pneumatic skeleton, which plays a crucial role in facilitating such movements. Moreover, when the pressure was increased to 1.5 bar, the manipulator exhibited a significantly greater axial displacement of about 2.35 N over the same distance, representing an increase by a factor of more than 3.6 times compared to the displacement observed at 0.9 bar. Through modulation of internal pneumatic pressure, the soft manipulator exhibits adjustable stiffness across all tested axes, significantly enhancing its ability to suppress exterior disturbance and perform tasks under load. Experimental findings reveal an expanded stiffness adjustment range in the vertical downward direction, indicating potential effectiveness in suspended gripping tasks. The qualities render the proposed design well-suited for suspension manipulations and the development of mechanical grippers with adjustable workspace and stiffness.

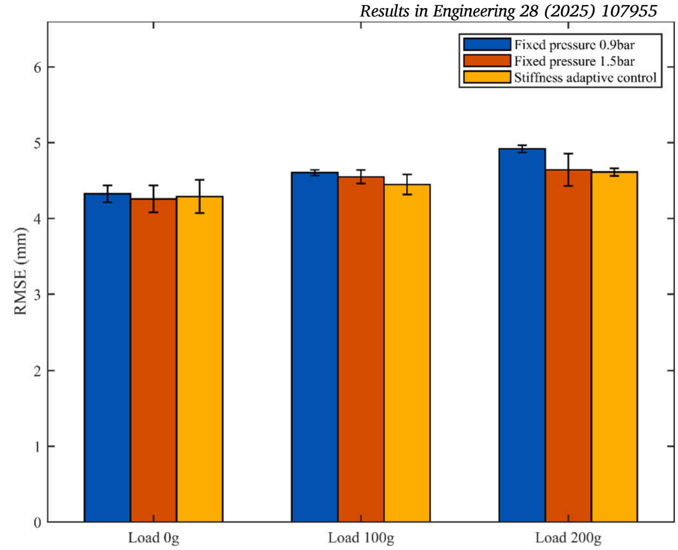

Fig. 11. Mean squared error of different control methods.

6.3. Tip trajectory tracking with loading

The spatially continuous trajectory is discretized into a series of distinct spatial points. The manipulator is controlled to track the tip's trajectory through simultaneous optimization of its configuration and stiffness properties. These properties are dynamically modified in response to external forces, enhancing functional adaptability and efficiency. This improves the precision of movements and the ability to adapt to varying loads.

Experiments monitored spatial spiral tracking under 0 g, 100 g, and 200 g load scenarios, as shown in Fig. 10(a). The external load at the manipulator's distal end induced a marked discrepancy between actual and designed paths, with deviation magnitude scaling proportionally with load weight. Implementing variable stiffness control allows the robot to autonomously regulate internal air pressure to accommodate varying external loads and maintain fidelity to diverse reference trajectories. As depicted in Fig. 10(b), soft robot operational air pressure escalates with increasing load, attributed to augmented trajectory deviation under load. For instance, under a 200 g load, the average operational air pressure was 1.02 bar, representing a 31.83% reduction relative to the fixed 1.5 bar setting. This reduction coincided with a decreased mean-square error (Fig. 11), showing improved tracking accuracy. However, continuous air pressure modulation may introduce minor oscillations and fluctuations in tracking error, potentially impairing overall performance. This highlights the need for pressure adjustment refinement in future research. The proposed methodology facilitates dynamic stiffness modification across trajectories, independent of specific end load. The experiment results demonstrate the effectiveness of the method in following diverse spatial trajectories under varying loads, validating system effectiveness. The research delineates a novel approach to enhancing soft robot adaptability and accuracy in trajectory-tracking applications, particularly under variable loading.

6.4. Application

The hybrid-driven soft robot described in this study exhibits inherent flexibility, ensuring safe operation, while dynamically adjusting stiffness to accommodate varying external loads and environmental disturbances. It has a large workspace and robust load-bearing capabilities, enabling efficient manipulation and placement of objects. In the experimental setup, a vacuum suction cup with a 20 mm diameter is attached to the end segment of the soft manipulator to facilitate object adhesion. This is achieved by employing a vacuum pump (Kamoer KLVP1-SB24L) to create negative pressure. Fig. 12 illustrates the variations in end po-

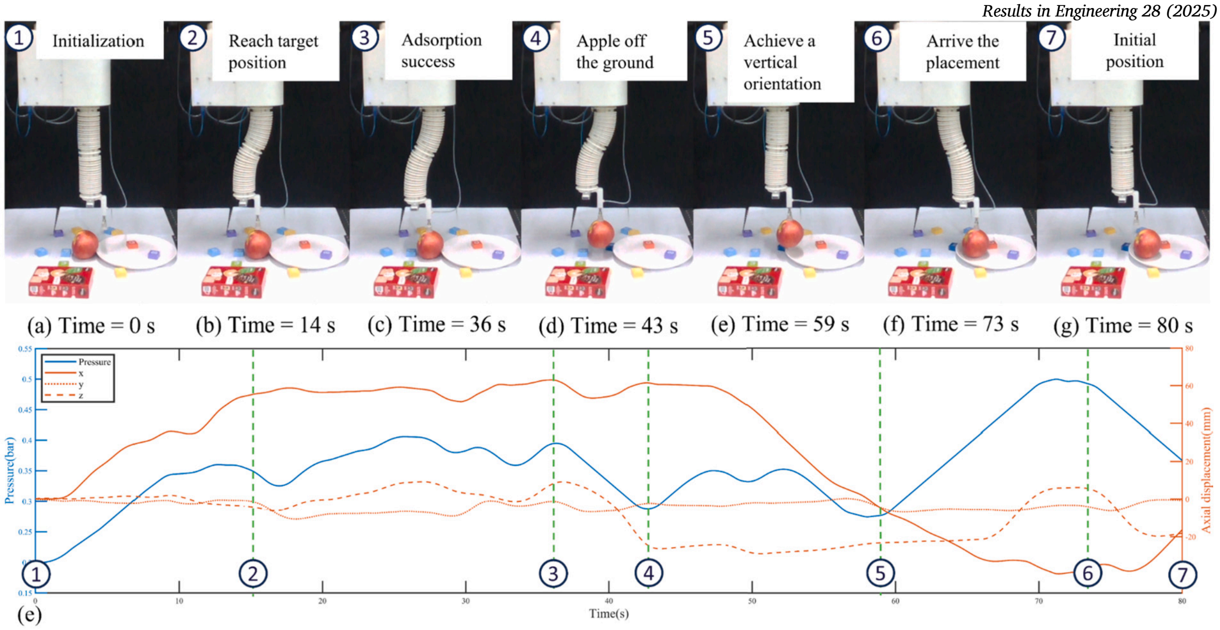

Fig. 12. Utilize the active stiffness control capability of the hybrid-driven soft manipulator to achieve object handling and placement tasks.

sition and internal air pressure observed during the actuation of the soft manipulator as it secures a 270 g apple.

In the exploratory experiment of the apple manipulation task, the procedure commences at time = 0 s. Initially, the soft manipulator is in its default state, as depicted in Fig. 12(a). Subsequently, the distal end of the robot is directed to move above the apple. To ensure structural stability during this movement, the air pressure is gradually escalated. By time = 14 s, the manipulator achieves the desired positioning, following which a command is issued to lower the manipulator, facilitating the suction cups' contact with the apple's surface (Fig. 12(b)). Simultaneously, the manipulator fine-tunes the air pressure upwards to enhance the physical stiffness. This adjustment is pivotal in counteracting the deformation induced by the upward force exerted by the spring within the suction cup. Nevertheless, variations in air pressure and positioning are observed, attributed to the oscillation of the apple. Upon successful adhesion at time = 36 s, a lifting directive is executed (Fig. 12(c)). At first, the air pressure decreases as the anticipated motion aligns with the spring force, promoting upward movement. By time = 43 s, the apple leaves the ground, and the air pressure is augmented to offset gravitational forces, as depicted in Fig. 12(d). Because deformation due to gravitational forces aligned with the intended direction of motion, there was a notable reduction in air pressure until the soft robot achieved a vertical orientation at time = 59 s (Fig. 12(e)). The control system then manipulated the arm to align with the designated target location. Before reaching this position, there was an increment in air pressure, strategically implemented to mitigate the gravitational influence of the apple. The final phase unfolds at time = 73 s, where the control system instructs the manipulator to revert to its initial position, as shown in Fig. 12(f). The air pressure significantly decreases due to the absence of the apple's weight and the reduced minimum air pressure necessary for structural stability. The operational cycle culminates as the robot returns to its starting configuration. This sequence elucidates the intricate interplay between air pressure modulation, gravitational influences, and the preservation of structural stability during the soft robot's interaction with the apple, emphasizing the pivotal role of stiffness control in robotic manipulation tasks.

7. Conclusion

This study has developed a hybrid-driven soft manipulator that integrates tendon and pneumatic actuation to achieve a large workspace and continuous stiffness modulation. Beyond the hardware, our primary contribution is a novel control framework that enables the robot to simultaneously and adaptively regulate its configuration and stiffness under external loads.

Our work provides three key contributions. First, we established a robust kinematic model by fusing the Piecewise Constant Curvature (PCC) model with a Weighted Broad Learning System (WBLS), which accurately captures the nonlinear relationship between configuration and the minimum driving pressure while being resilient to experimental noise. Second, we proposed an adaptive optimization-based control strategy. This method co-optimizes configuration and stiffness in its inverse kinematics solution and incorporates an online stiffness estimator that uses end-effector feedback to autonomously counteract load-induced deformations in real-time. Finally, experimental validation demonstrated that the robot achieves a wide adjustable range of axial stiffness, with more than a 3.6-fold increase, along with improvements in control accuracy. Under a 200 g load, our method improved trajectory tracking accuracy while achieving up to a 31.83% reduction in average driving pressure, and its practical efficacy was demonstrated through a successful object manipulation task.

Despite these results, limitations from modeling inaccuracies and pneumatic hysteresis can lead to slight oscillations and suboptimal stiffness estimation. Future work will therefore focus on establishing a high-fidelity simulation environment for soft robots and subsequently exploring simulation-to-real-world transfer learning to further enhance performance in complex scenarios. These advances position our framework as a foundation for next-generation adaptive soft robots in human-robot interaction applications.

CRediT authorship contribution statement

Xin Fu: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Daohui Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology. **Shuheng Ren:** Visualization, Validation, Formal analysis. **Yaqi Chu:** Writing – review & editing, Validation, Supervision. **Xingang Zhao:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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